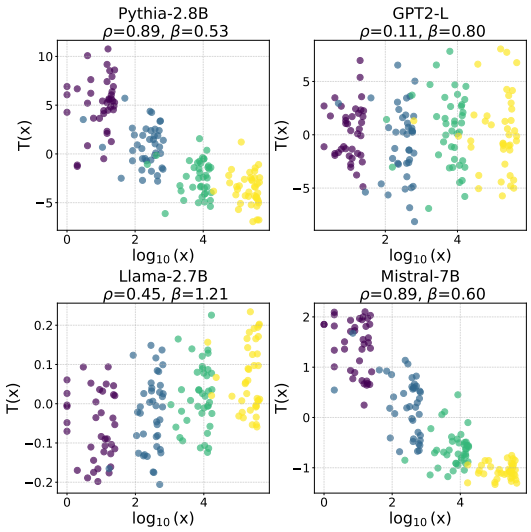

Figure 5: Projections of letters representations (y-axis) against their log-scaled magnitudes (x-axis) assigned proportional to their length, for the layer with the highest explained variance in four models. Sublinearity and monotonicity (ρ) are indicated above each subfigure, demonstrating consistent sublinear trends and strong monotonic relationships across models.

ers with comparable σ^2 values may exhibit similar trends. Figure 3 provides a layer-wise analysis for four models, demonstrating how sublinearity evolves across different depths.

PCA vs PLS. The PLS method achieves high monotonicity (ρ) and explained variance (R^2) but exhibits lower sublinearity compared to PCA. This discrepancy arises because PLS operates as a supervised linear probe, where the regression target (e.g., numerical values) directly influences the projection. This process distorts the intrinsic spacing between points, as PLS prioritizes maximizing covariance with the target over preserving the original geometric structure. In contrast, PCA, being unsupervised, retains the relative spacing of data points in the latent space, better capturing the underlying sublinear trends. This distinction is evident in tables 1 and 2: PCA consistently reveals stronger sublinearity, while PLS achieves higher R^2 and ρ by aligning the projection with the target variable.

Notably, this aligns with findings in Zhu et al. (2025), where a linear probe failed to adequately capture the non-linear scaling of hidden states, particularly for larger numbers, where non-linearity becomes more pronounced. Our work explicitly quantify sublinearity using the Scaling Rate Index (SRI, β), which directly measures the rate of scaling in the latent space. This allows us to better capture the true geometric organization of numerical representations, especially in regimes where non-linear effects dominate.

Ablation study. Finally, we perform an ablation study to examine the dependence of our results on the number of examples in the prompt for both numerical and alphabetical datasets. Figure 6 shows that the metrics exhibit greater stability for numerical data compared to alphabetical data, indicating that the model processes numerical information more consistently, while alphabetical representations are more sensitive to prompt variations.

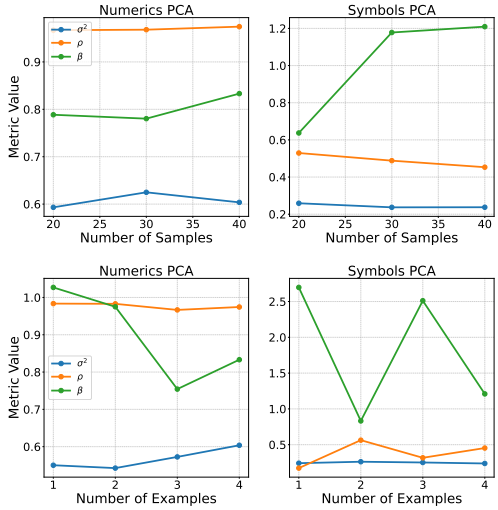

Figure 6: Top row: Change in metrics with respect to the number of samples in the set $\mathcal{X}$. Bottom row: Change in metrics with respect to the number of in-context examples in the prompt. Left column corresponds to the **numbers** group, and the right column corresponds to the **letters** group. Sublinearity and monotonicity trends are highlighted for each case.

5 Experiment 2: Identifying number line using real-world tasks

Setup. In the previous experiment (Section 4) we created an artificial experimental setting to test our hypothesis. In this experiment, however, we want to further validate our hypothesis using real-world data. We collect names of celebrities along with their birth years and population of different cities/-countries from Wikidata (Vrandečić and Krötzsch, 2014). The task here is to investigate for similar patterns and observations seen in the previous experiment.

Motivation. The goal of this experiment is to investigate how LLMs internally represent numerical

values in real-world contexts, specifically focusing on the monotonicity and scaling of these representations. By analyzing the hidden states, we aim to uncover whether the models encode numerical information in a structured and interpretable manner.

Methodology. The experimental setup for this experiment is as follows:

• **Prompting the Model:** We prompt the model to provide the exact birth year or population size for each entity in our dataset, which consists of 1K samples.An example of a prompt would be "What is the population of [country]?" • **Collecting Model Outputs:** We collect the LLM's output answers, and filter out non-numerical and incorrect responses.

• **Extracting Hidden States:** We extract the hidden state corresponding to the question mark token at each model layer[5].

• **Training PLS Models:** We train one- and two-component PLS models on the extracted hidden states to predict the birth years or population sizes of the entities. This is performed for two LLMs: *Llama-3.1-8B* and *Llama-3.2-1B*.

Results. The results, as shown in table 3, demonstrate a clear distinction between the two tasks, but also between the models.

For the *birth year* task, model *Llama-3.1-8B* exhibit strong trends, with high monotonicity (ρ) and (R^2), while having low SRI (β), hence high compression. This indicates that the internal representations of birth years are well-structured and predictive, aligning with our expectations for numerical encoding in LLMs. On the other side, *Llama-3.2-1B*) shows low monotonicity score, hence the β factor is not informative. We attribute the low monotonicity score to the non-structured internal representations of lower birth years, as can be seen in Figure 7.

For the *population size* task, both models display weaker monotonicity and lower R^2, suggesting population sizes are encoded less systematically. Unlike birth years, population figures are more context-dependent, influenced by geopolitical changes, reporting inconsistencies, and approximate expressions in text. Consequently, the low monotonicity makes the scaling ratio β unreliable.

Finally Figure 7 shows the examples of one and two PLS projections for two models, for birth-year dataset.

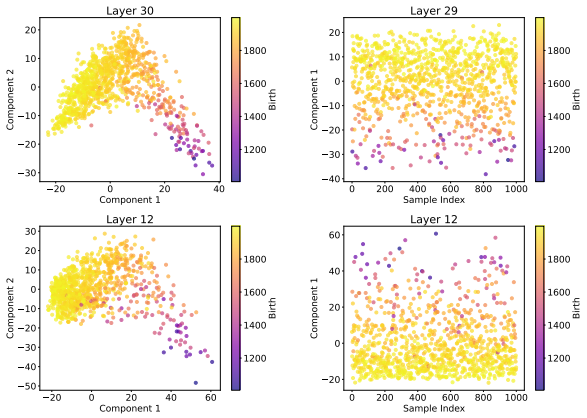

Figure 7: Visualization of PLS models trained on Llama-3.1-8B (top row) and Llama-3.2-1B (bottom row) model activations to predict entities' birth years using one and two dimensional PLS respectively. Each subfigure represents the layer with the highest R^2 score for one- and two-component PLS models.

Model	Dataset	Layer	ρ	β	R^2
Llamba 3.1 8B	Birth	29	0.84	0.50	0.63
	Population	9	0.63	2.48	0.08
Llamba 3.2 1B	Birth	12	0.03	0.72	0.61
	Population	10	0.62	2.69	0.10

Table 3: Results for Llamba models evaluated on the Birth and Population datasets. Results are reported for the layer with the highest R^2, highlighting the relationship between scaling rate (β), monotonicity (ρ), and model performance.

6 Conclusion

Inspired by the logarithmic compression in human numerical cognition, we investigate whether LLMs encode numerical values analogously. By analyzing hidden states across layers, we employ dimensionality reduction techniques (PCA and PLS) and geometric regression to test for two key properties: (1) order preservation and (2) sublinear compression, where distances between consecutive numbers decrease as values increase. Our results reveal that while both PCA and PLS identify numerical representations in a linear subspace, only PCA captures systematic sublinearity. This indicates that linear probes like PLS, which optimize for covariance with the target, may obscure the underlying non-uniform structure. Our findings suggest that LLMs encode numerical values with structured compression, akin to the human mental number line, but

[5]We divided the hidden states into four equally sized groups, ranging from the minimum to the maximum answers, to facilitate the calculation of the Scaling Rate Index (SRI).

this is only detectable through methods like PCA that preserve geometric relationships.

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